

Deep Reinforcement Learning → Power Electronic Converter System

Model-Based

Alpha Zero	MBMF	MBVE	I2A	World-Models
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Model-Free

Value-based	Policy-based	Actor-Critic	Multi-Agent
Q-Learning	REINFORCE	DPG	MADDPG
SARSA	SVPG	DDPG	MAPPO
DQN	TRPO	A3C	COMA
Double DQN	PPO	ACKTR	MASAC
Dueling DQN	PPG	ACER	QMIX
DRQN	GRPO	SAC	HAPPO
DD-DQN	KL-A PPO	IMPALA	VDN

Design
(Section III)

Control
(Section IV)

Maintenance
(Section V)

Topology Generator

Parameter optimization

Maximum power point tracking

Modulation

Variable-Parameter Tuning

Energy management

Motor Speed Track

Predictive Maintenance

Thermal Reliability aware

Three Life-cycle Phase of Power Electronic Converter systems